

Prediction-Powered Data Fusion for Treatment Effect Estimation

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Randomized controlled trials identify causal effects under transparent design assumptions, but their sample sizes can make average and conditional treatment-effect estimation imprecise. Observational studies are often larger and can yield accurate outcome predictions, but those predictions need not be causal. We develop a model-agnostic prediction-powered framework that keeps the randomized trial as the causal anchor while using observational outcome models and covariates through two tunable channels. The coefficient λ changes the outcome regression inside an RCT-valid pseudo-outcome, and ω introduces a prediction-powered control variate centered across the two samples.

For the average treatment effect, we derive exact conditional unbiasedness, an exact two-sample variance formula, the joint population variance minimizer, its population-orthogonal simplification, and an honest plug-in algorithm. For conditional effects, we formulate a prediction-powered squared-loss learner from an RCT-valid AIPW pseudo-outcome, distinguish target preservation from empirical loss-variance tuning, and give implementable linear-sieve and frozen-neural-basis versions. We outline an R-loss analogue only as a prospective extension requiring a separate derivation. Exact density-ratio transport covers unequal covariate marginals, while loss-targeted balancing gives an approximate alternative with explicit drift. Existing simulations support the ATE construction in several small-trial regimes. CATE experiments and real-outcome analyses remain planned, so the present empirical evidence does not establish a universal finite-sample improvement guarantee.

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1. Introduction

1.1. Categorized literature review

Prediction-powered inference. Prediction-powered inference uses a large set of machine predictions together with a smaller labeled sample, correcting the prediction-based estimate with labeled residuals so that the statistical target does not require the predictor to be correct (Angelopoulos et al., 2023). This separation between predictive quality and inferential validity is the starting point for the present construction. Randomized observations play the role of causally valid labeled data, while observational outcome models supply potentially useful predictions.

Adaptive prediction-powered inference. PPI++ introduces data-adaptive power tuning that adjusts how strongly predictions enter an estimating equation and reports computational and statistical gains over the original construction (Angelopoulos et al., 2024). Its central lesson is that a prediction should be used in proportion to its empirical value rather than being accepted as truth. Our coefficient ω follows this control-variate logic, but the target and score are causal quantities identified by the randomized design.

Causal data fusion and transport. General causal data-fusion theory distinguishes data collected under different regimes and populations and asks whether a target causal query is identified from their combination (Bareinboim and Pearl, 2016). Reviews of causal data integration likewise separate randomized trials, observational studies, historical controls, and heterogeneous target populations (Shi et al., 2023). The present paper assumes that the trial already identifies the target effect and studies an estimation question: how can observational predictions improve precision without replacing randomized identifying information?

Combining randomized and observational estimates. Existing estimators combine experimental and observational effect estimates through shrinkage, bias detection, or adaptive weighting. Shrinkage estimators can reduce risk when observational bias is sufficiently controlled (Rosenman et al., 2023), while adaptive combination methods move toward the randomized estimator when observational bias is detected (Cheng and Cai, 2021). These approaches combine effect estimates from the two sources. Our construction instead inserts the observational outcome model into an RCT-valid pseudo-outcome and separately centers a prediction-based covariate statistic.

CATE learners and orthogonal losses. Meta-learners convert standard regression algorithms into conditional effect estimators (Künzel et al., 2019). The R-learner residualizes outcomes and treatments to isolate treatment-effect heterogeneity (Nie and Wager, 2021), while doubly robust pseudo-outcome regression can retain favorable error structure under nuisance estimation (Kennedy, 2023). Direct multi-source learners have also been proposed for heterogeneous causal data fusion (Li et al., 2022). These results motivate treating the prediction-powered construction as a loss wrapper that can be paired with a linear sieve or a fixed learned representation.

Covariate shift and balancing. When trial and observational covariate marginals differ, exact density-ratio weighting transports fixed covariate functions between populations. Riesz-representer methods provide an optimization perspective on such linear functionals and on debiasing (Chernozhukov et al., 2021). Density-ratio estimation, generic balancing, and Riesz learning are established tools. The project-specific question is whether they can be targeted to the generated CATE loss class $\{(g - t)^2 : t \in \mathcal{T}\}$ while preserving an RCT-defined risk; we treat this specialization as a working research direction rather than a settled novelty claim.

2. Background and Problem Setting

2.1. Prediction-powered inference and PPI++

Let (X_i, Y_i) , $i = 1, \dots, n$, be labeled pairs and let $\tilde{X}_j$, $j = 1, \dots, N$, be unlabeled covariates. Suppose a fixed predictor supplies $f(X_i)$ and $f(\tilde{X}_j)$. For a mean target, the elementary prediction-powered decomposition is

$$\hat{\mu}_{\text{PPI}} = \frac{1}{N} \sum_{j=1}^N f(\tilde{X}_j) + \frac{1}{n} \sum_{i=1}^n \{Y_i - f(X_i)\}. \quad (1)$$

When labeled and unlabeled covariates share a marginal law and the two samples are independent, the prediction term and residual correction sum to the labeled-population mean for every fixed predictor f . PPI++ generalizes this idea by tuning the strength of the prediction contribution. In our construction, ω is the PPI-style scale on a mean-zero prediction control variate. By contrast, λ blends trial and observational outcome regressions inside an RCT-valid score; it is not the scale on the external prediction mean.

2.2. RCT–OBS problem setting and notation

We observe independent samples

$$D_R = \{V_i^R = (X_i^R, A_i^R, Y_i^R) : i = 1, \dots, n\} \sim P_R^{\otimes n}, \quad (2)$$

$$D_O = \{V_j^O = (X_j^O, A_j^O, Y_j^O) : j = 1, \dots, N\} \sim P_O^{\otimes N}, \quad (3)$$

where X is a covariate vector, $A \in \{0, 1\}$ is treatment, and Y is the observed outcome. For a function h , write $\mathbb{P}_{R,n}h = n^{-1} \sum_i h(V_i^R)$ and $\mathbb{P}_{O,N}h = N^{-1} \sum_j h(V_j^O)$.

Let $Y(1)$ and $Y(0)$ denote potential outcomes. The RCT-population conditional and average treatment effects are

$$\tau(x) = \mathbb{E}_R\{Y(1) - Y(0) \mid X = x\}, \quad \theta = \mathbb{E}_R\{\tau(X)\}. \quad (4)$$

Assumption 1 (Trial identification). *Consistency holds, $\{Y(0), Y(1)\} \perp A \mid X$ under P_R , and the known trial propensity $e(x) = P_R(A = 1 \mid X = x)$ satisfies $\epsilon < e(X) < 1 - \epsilon$ almost surely for a constant $\epsilon > 0$.*

Assumption 2 (Covariate relation). *The common-marginal model assumes $P_R^X = P_O^X$. The covariate-shift extension assumes $P_R^X \neq P_O^X$, $P_R^X \ll P_O^X$, and uses the exact density ratio*

$$r(x) = \frac{dP_R^X}{dP_O^X}(x), \quad \mathbb{E}_O\{r(X)h(X)\} = \mathbb{E}_R\{h(X)\} \quad (5)$$

for every required integrable function h . The common-marginal case is the special case $r = 1$.

All exact finite-sample statements use honest conditioning. Nuisance functions and adaptive coef-

ficients are trained on blocks independent of the final evaluation observations. Conditional on the resulting pre-evaluation sigma-field $\mathcal{F}$, the fitted objects are fixed and the two evaluation samples retain the product law $P_R^{\otimes n} \otimes P_O^{\otimes N}$.

Let $\hat{\mu}_R = (\hat{\mu}_{R,0}, \hat{\mu}_{R,1})$ and $\hat{\mu}_O = (\hat{\mu}_{O,0}, \hat{\mu}_{O,1})$ be outcome regressions fitted on the two sources. Define

$$g(x) = \hat{\mu}_{O,1}(x) - \hat{\mu}_{O,0}(x), \quad (6)$$

$$\varphi(V; q) = q_1(X) - q_0(X) + \frac{A}{e(X)} \{Y - q_1(X)\} - \frac{1 - A}{1 - e(X)} \{Y - q_0(X)\}. \quad (7)$$

For every fixed q , Assumption 1 gives

$$\mathbb{E}_R\{\varphi(V; q) \mid X\} = \tau(X). \quad (8)$$

For $\lambda \in \mathbb{R}$, set

$$\hat{\mu}_\lambda = (1 - \lambda)\hat{\mu}_R + \lambda\hat{\mu}_O, \quad Z_\lambda = \varphi(V; \hat{\mu}_\lambda) = Z_0 + \lambda\Delta, \quad \Delta = Z_1 - Z_0. \quad (9)$$

Equation (8) implies $\mathbb{E}_R(Z_\lambda \mid X) = \tau(X)$ and $\mathbb{E}_R(\Delta \mid X) = 0$. The observational contrast g is used only as a prediction and need not equal τ .

2.3. Existing data-fusion estimators

Three existing strategies provide the nearest comparisons. First, shrinkage methods combine an unbiased experimental estimate with a potentially biased observational estimate and trade variance against observational bias (Rosenman et al., 2023). Second, adaptive RCT-OBS estimators estimate source-specific conditional effects and downweight the observational estimator when its discrepancy from the trial is large (Cheng and Cai, 2021). Third, direct multi-source learners construct weighted pseudo-outcomes for heterogeneous effects across sources (Li et al., 2022). Our estimator differs at the level of the fusion object: λ changes the nuisance regression inside a trial-valid score, while ω centers a fixed prediction across study covariate samples. Neither channel treats the observational treatment contrast as identified causal truth.

3. Prediction-Powered Fusion for the ATE

3.1. Estimator and exact variance

For fixed (λ, ω) , define

$$\hat{\theta}_r(\lambda, \omega) = \mathbb{P}_{R,n}Z_\lambda + \omega \{ \mathbb{P}_{O,N}\{r(X)g(X)\} - \mathbb{P}_{R,n}g(X) \}. \quad (10)$$

The first term averages an RCT-valid pseudo-outcome. The expression in braces is a mean-zero control variate under Assumption 2. Thus, the RCT identifies the target, λ changes within-covariate score noise, and ω uses the large OBS covariate sample to remove predictable between-covariate variation.

Theorem 1 (Exact conditional mean and variance). *Under Assumptions 1 and 2, honest conditioning, and square integrability of the displayed scores, every fixed (λ, ω) satisfies*

$$\mathbb{E}\{\hat{\theta}_r(\lambda, \omega) \mid \mathcal{F}\} = \theta, \quad (11)$$

$$\text{Var}\{\hat{\theta}_r(\lambda, \omega) \mid \mathcal{F}\} = \frac{1}{n} \text{Var}_R\{Z_\lambda - \omega g(X)\} + \frac{\omega^2}{N} \text{Var}_O\{r(X)g(X)\}. \quad (12)$$

The mean follows from $\mathbb{E}_R(Z_\lambda) = \theta$ and $\mathbb{E}_O(rg) = \mathbb{E}_R(g)$. The variance follows after rewriting Equation (10) as the sum of independent RCT and OBS empirical means.

3.2. Joint and population-orthogonal oracles

Define the five conditional population moments

$$A = \text{Var}_R(\Delta), \quad B = \text{Var}_R(g) + \frac{n}{N} \text{Var}_O(rg), \quad (13)$$

$$C = \text{Cov}_R(Z_0, \Delta), \quad D = \text{Cov}_R(Z_0, g), \quad E = \text{Cov}_R(\Delta, g). \quad (14)$$

With

$$\mathbf{H} = \begin{pmatrix} A & -E \\ -E & B \end{pmatrix}, \quad \ell = \begin{pmatrix} C \\ -D \end{pmatrix}, \quad \beta = \begin{pmatrix} \lambda \\ \omega \end{pmatrix}, \quad (15)$$

the compact variance in Equation (12) becomes

$$\text{Var}\{\hat{\theta}_r(\beta) \mid \mathcal{F}\} = \frac{1}{n} \left\{ \text{Var}_R(Z_0) + 2\ell^\top \beta + \beta^\top \mathbf{H} \beta \right\}. \quad (16)$$

The matrix $\mathbf{H}$ is positive semidefinite because

$$\begin{pmatrix} u & v \end{pmatrix} \mathbf{H} \begin{pmatrix} u \\ v \end{pmatrix} = \text{Var}_R(u\Delta - vg) + \frac{n}{N} v^2 \text{Var}_O(rg) \geq 0. \quad (17)$$

Theorem 2 (Joint population oracle). *If $AB - E^2 > 0$, the unique unrestricted minimizer of Equation (16) is*

$$\lambda_{\text{joint}}^* = \frac{ED - BC}{AB - E^2}, \quad \omega_{\text{joint}}^* = \frac{AD - EC}{AB - E^2}. \quad (18)$$

If $\mathbf{H}$ is singular, the minimizer set is $-\mathbf{H}^+ \ell + \text{Null}(\mathbf{H})$. For a prespecified compact feasible set, a constrained minimizer exists and is obtained by minimizing the same quadratic.

For $\rho > 0$, a ridge-stabilized population coefficient is

$$\beta_\rho^* = -(\mathbf{H} + \rho I_2)^{-1} \ell. \quad (19)$$

It minimizes the penalized quadratic $2\ell^\top \beta + \beta^\top \mathbf{H} \beta + \rho \|\beta\|_2^2$. This is a penalized stability oracle, not the oracle for the unpenalized variance in Equation (16).

The conditional identity $\mathbb{E}_R(\Delta \mid X) = 0$ gives the exact population orthogonality

$$E = \text{Cov}_R\{\Delta, g(X)\} = 0. \quad (20)$$

It also gives $D = \text{Cov}_R\{\tau(X), g(X)\}$. When $A, B > 0$, the population oracle therefore separates:

$$\lambda^* = -\frac{C}{A}, \quad \omega^* = \frac{D}{B}. \quad (21)$$

The population oracle weakly improves the variance objective whenever $(0, 0)$ is feasible. Estimated coefficients have no universal finite-sample no-harm guarantee.

3.3. Estimable honest algorithm

Let $s_R(h_1, h_2)$ and $s_O(h_1, h_2)$ denote unbiased sample covariances on independent tuning blocks, with $s_S^2(h) = s_S(h, h)$. Compute

$$\hat{A} = s_R^2(\Delta), \quad \hat{B} = s_R^2(g) + \frac{n}{N} s_O^2(rg), \quad (22)$$

$$\hat{C} = s_R(Z_0, \Delta), \quad \hat{D} = s_R(Z_0, g), \quad \hat{E} = s_R(\Delta, g). \quad (23)$$

The population-orthogonality plug-in imposes $\hat{E} = 0$ and uses $(-\hat{C}/\hat{A}, \hat{D}/\hat{B})$. The empirical joint minimizer retains $\hat{E}$ and applies Equation (18). The two rules differ because sample orthogonality is not exact.

Algorithm 1 states the exact three-way implementation.

Algorithm 1: Three-way honest ATE estimation

Input: Independent RCT and OBS samples; a prespecified feasible box and denominator thresholds.

- 1 **Split:** Divide each source into nuisance-training, coefficient-tuning, and final-evaluation blocks.
- 2 **Fit:** Estimate $\hat{\mu}_R$, $\hat{\mu}_O$, g , and, if required, a transport model without using any tuning or evaluation observations. The exact theorem uses the true ratio r ; replacing it by $\hat{r}$ defines the approximate regime described below.
- 3 **Tune:** On the independent tuning blocks, compute the five moments, apply the prespecified denominator thresholds, and solve the box-constrained quadratic. Use $(0, 0)$ as the declared fallback when the moments are unstable.
- 4 **Evaluate:** Hold every fitted object fixed and evaluate Equation (10) on the untouched final blocks.
- 5 **Infer:** Set $\psi_R = Z_{\hat{\lambda}} - \hat{\omega}g$ and $\psi_O = \hat{\omega}rg$, and report $\hat{V} = s_R^2(\psi_R)/n + s_O^2(\psi_O)/N$. Under conditional Lindeberg conditions, bounded conditional $(2 + \delta)$ moments, and a nondegenerate variance limit, use $\hat{\theta}_r \pm z_{1-\alpha/2} \sqrt{\hat{V}}$.

Output: $\hat{\theta}_r$, $\hat{V}$, and the corresponding Wald interval.

With an estimated ratio $\hat{r}$, define

$$h_n(X) = \{\hat{r}(X) - r(X)\}g(X). \quad (24)$$

The exact conditional drift is

$$\omega b_{\mathcal{F}} = \omega \mathbb{E}_O\{h_n(X)\}. \quad (25)$$

Honest splitting prevents evaluation leakage but does not remove this drift. A direct sufficient negligible-error regime is

$$\sqrt{n}\omega \mathbb{E}_O\{h_n(X)\} = o_p(1), \quad |\omega| \sqrt{\frac{n}{N}} \|h_n - \mathbb{E}_O(h_n)\|_{L^2(P_O)} = o_p(1). \quad (26)$$

Evaluation-only variance is first-order adequate under these conditions and the preceding CLT assumptions. Otherwise the ratio learner’s influence-function contribution, the OBS empirical-process term, and their covariances must be included.

4. Prediction-Powered Fusion Learning for CATE

4.1. Prediction-powered loss, coefficient roles, and the ATE relation

For a fixed candidate CATE function t , define the density-ratio-weighted prediction-powered squared loss

$$\widehat{L}_{\text{DR}}(t; \lambda, \omega) = \mathbb{P}_{R,n}\{Z_\lambda - t(X)\}^2 \quad (27)$$

$$+ \omega \left[\mathbb{P}_{O,N}\{r(X)(g(X) - t(X))^2\} - \mathbb{P}_{R,n}\{(g(X) - t(X))^2\} \right]. \quad (28)$$

For every fixed candidate considered below, assume $Z_\lambda, t \in L^2(P_R)$, and let $G_t(X) = \{g(X) - t(X)\}^2$ satisfy $G_t \in L^1(P_R)$ and $rG_t \in L^1(P_O)$. Under exact transport and honest conditioning,

$$\mathbb{E}\{\widehat{L}_{\text{DR}}(t; \lambda, \omega)\} = Q(\lambda) + \mathbb{E}_R\{t(X) - \tau(X)\}^2, \quad (29)$$

$$Q(\lambda) = \mathbb{E}_R\{Z_\lambda - \tau(X)\}^2. \quad (30)$$

The common-marginal case sets $r = 1$. In both cases, the expected loss is flat in ω . When $A > 0$, define the integrated pseudo-outcome-noise oracle

$$\lambda_Q^* = \arg \min_{\lambda \in \mathbb{R}} Q(\lambda) = -C/A. \quad (31)$$

Write $\lambda_{\text{ATE}}^* := \lambda^*$ and $\omega_{\text{ATE}}^* := \omega^*$ for the two coordinates of the population-orthogonal ATE variance oracle in Equation (21). Then $\lambda_Q^* = \lambda_{\text{ATE}}^*$. When $A = 0$, $\Delta = 0$ almost surely and $Q(\lambda)$ is flat in λ . The ATE variance coefficient $\omega_{\text{ATE}}^* = D/B$ is a possible training default, but Equation (30) does not identify it as a CATE-risk optimizer.

The sampling variance of a fixed candidate loss identifies a distinct coefficient. Let $U_{t,\lambda} = \{Z_\lambda - t(X)\}^2$. If the required fourth moments exist and the denominator is positive, then

$$\omega_{\text{LV}}^*(t; \lambda) = \frac{\text{Cov}_R(U_{t,\lambda}, G_t)}{\text{Var}_R(G_t) + (n/N) \text{Var}_O(rG_t)}. \quad (32)$$

This oracle minimizes the variance of one fixed empirical loss. It does not minimize the final learned CATE risk.

Final learner-risk oracle. Let $\mathcal{C}$ be a prespecified finite candidate class. An index

$$c = (\lambda_c, \omega_c, p_c, \rho_c) \in \mathcal{C} \quad (33)$$

specifies the two fusion coefficients, the sieve dimension, and the penalty level. All other components of the learning rule, including the data splits, nuisance algorithms, basis construction, and optimization convention, are also prespecified. Let $\mathcal{D}_{\text{tr}}$ collect all pre-validation randomness, including the candidate-training samples, nuisance and ratio training samples, basis fits, and independently randomized split assignments, and set $\mathcal{F}_{\text{train}} = \sigma(\mathcal{D}_{\text{tr}})$. Let $\hat{t}_c^{\mathcal{D}_{\text{tr}}}$ denote the resulting fitted function and define its sample-specific target risk and its algorithm-induced expected risk by

$$\mathcal{R}_{\mathcal{D}_{\text{tr}}}(c) = \mathbb{E}_R \left[\{ \hat{t}_c^{\mathcal{D}_{\text{tr}}}(X) - \tau(X) \}^2 \middle| \mathcal{D}_{\text{tr}} \right], \quad (34)$$

$$\mathfrak{R}(c) = \mathbb{E}_{\mathcal{D}_{\text{tr}}} \{ \mathcal{R}_{\mathcal{D}_{\text{tr}}}(c) \}. \quad (35)$$

The final CATE-risk oracle minimizes one of these risks over fitted learning algorithms. It is distinct from both λ_Q^* and $\omega_{\text{LV}}^*(t; \lambda)$.

To make the fallback comparison exact, let

$$\mathcal{C}_R = \{(0, 0, p, \rho) : (p, \rho) \in \mathcal{H}_R\} \subseteq \mathcal{C} \quad (36)$$

contain the identical RCT-only learners over the prespecified sieve and penalty grid $\mathcal{H}_R$. Identical means that the RCT observations, folds, RCT nuisance fits, basis, penalty scaling, and optimization code agree with the corresponding fusion candidates; only the two OBS fusion channels are set to zero.

Proposition 1 (Candidate-class oracle dominance). *For every realized candidate-training sample for which the displayed infima are finite,*

$$\inf_{c \in \mathcal{C}} \mathcal{R}_{\mathcal{D}_{\text{tr}}}(c) \leq \inf_{c \in \mathcal{C}_R} \mathcal{R}_{\mathcal{D}_{\text{tr}}}(c). \quad (37)$$

Likewise,

$$\inf_{c \in \mathcal{C}} \mathfrak{R}(c) \leq \inf_{c \in \mathcal{C}_R} \mathfrak{R}(c). \quad (38)$$

If the infima are attained, they define the sample-specific and expected-training final CATE-risk oracles. Either inequality is strict only when at least one fusion candidate has risk strictly below the optimally tuned RCT-only class under the corresponding comparison.

Proof. Both inequalities follow from the set inclusion $\mathcal{C}_R \subseteq \mathcal{C}$. Strict inclusion of candidate sets alone does not imply a strict risk inequality. $\square$

Proposition 1 is an infeasible oracle statement because its risks use the unknown τ . It does not say that a data-adaptive selector dominates the RCT-only learner on every finite sample.

Common-reference honest validation. Condition on the pre-validation sigma-field $\mathcal{F}_{\text{train}}$ defined above. Construct one shared RCT-valid pseudo-outcome $Z_{\lambda_{\text{ref}}}$ on an untouched RCT validation

block using a fixed reference coefficient. For candidate c , set

$$G_c(X) = \{g(X) - \hat{t}_c^{\mathcal{D}_{\text{tr}}}(X)\}^2. \quad (39)$$

Let $r^\dagger = 1$ under the common covariate marginal, $r^\dagger = r$ under exact covariate transport, and $r^\dagger = \hat{r}$ in the estimated-ratio regime. Define the shared-reference validation score

$$\begin{aligned} \hat{\mathcal{R}}_c^{\text{ref}} &= \mathbb{P}_{R,\text{val}} \left[\{Z_{\lambda_{\text{ref}}} - \hat{t}_c^{\mathcal{D}_{\text{tr}}}(X)\}^2 \right] \\ &\quad + \mathbb{P}_{O,\text{val}} \{r^\dagger(X)G_c(X)\} - \mathbb{P}_{R,\text{val}} G_c(X). \end{aligned} \quad (40)$$

All candidates must use the same reference pseudo-outcome, g , ratio rule, and validation observations. Under the common marginal or exact r , known RCT propensity, and honest conditioning,

$$\mathbb{E}\{\hat{\mathcal{R}}_c^{\text{ref}} \mid \mathcal{F}_{\text{train}}\} = Q(\lambda_{\text{ref}}) + \mathcal{R}_{\mathcal{D}_{\text{tr}}}(c). \quad (41)$$

Thus candidate comparisons remove the same additive reference noise. The OBS correction is not required for validity of this identity, but it can reduce the sampling variability of validation when the prediction is useful.

For the approximate regime, define the conditional reference bias

$$b_{\text{ref}}(X) = \mathbb{E}_R \{Z_{\lambda_{\text{ref}}} - \tau(X) \mid X, \mathcal{F}_{\text{train}}\}, \quad (42)$$

the ratio and nuisance drifts

$$\delta_{r,c} = \mathbb{E}_O [\{r^\dagger(X) - r(X)\}G_c(X)], \quad (43)$$

$$\delta_{\text{nuis},c} = 2\mathbb{E}_R \left[b_{\text{ref}}(X) \{\tau(X) - \hat{t}_c^{\mathcal{D}_{\text{tr}}}(X)\} \right], \quad (44)$$

and their uniform magnitudes

$$\Delta_r = \sup_{c \in \mathcal{C}} |\delta_{r,c}|, \quad \Delta_{\text{nuis}} = \sup_{c \in \mathcal{C}} |\delta_{\text{nuis},c}|. \quad (45)$$

Known trial propensity and an honest reference nuisance give $b_{\text{ref}} = 0$ for every outcome-regression fit. Exact transport gives $\Delta_r = 0$. Estimated ratios or estimated propensities must instead be controlled through the displayed drifts or a stronger problem-specific expansion. Adaptive reuse of validation observations violates the held-out product law and measurability used here; it is outside Proposition 2 unless a separate selective-inference or stability theorem is supplied.

Proposition 2 (Validation-selector oracle inequality). *Let*

$$\hat{c} \in \arg \min_{c \in \mathcal{C}} \hat{\mathcal{R}}_c^{\text{ref}} \quad (46)$$

and define the conditional validation fluctuation

$$\varepsilon_{\text{val}} = \sup_{c \in \mathcal{C}} \left| \hat{\mathcal{R}}_c^{\text{ref}} - \mathbb{E}\{\hat{\mathcal{R}}_c^{\text{ref}} \mid \mathcal{F}_{\text{train}}\} \right|. \quad (47)$$

Then

$$\mathcal{R}_{\mathcal{D}_{\text{tr}}}(\hat{c}) \leq \inf_{c \in \mathcal{C}} \mathcal{R}_{\mathcal{D}_{\text{tr}}}(c) + 2\varepsilon_{\text{val}} + 2\Delta_r + 2\Delta_{\text{nuis}} \quad (48)$$

and therefore the same upper bound holds with the infimum over $\mathcal{C}$ replaced by the larger infimum over $\mathcal{C}_R$.

Proof. Conditionally on $\mathcal{F}_{\text{train}}$, expand the expectation of Equation (40). The squared pseudo-outcome term equals the target risk plus $Q(\lambda_{\text{ref}})$ and $\delta_{\text{nuis},c}$; the transported correction contributes $\delta_{r,c}$. Compare the empirical score of $\hat{c}$ with that of any $c \in \mathcal{C}$, bound the two empirical deviations by ε_{val} , and bound the two drift differences by $2\Delta_r + 2\Delta_{\text{nuis}}$. Taking the infimum proves the result. $\square$

If $\varepsilon_{\text{val}} + \Delta_r + \Delta_{\text{nuis}} = o_p(1)$, Proposition 2 gives asymptotic weak noninferiority relative to the identical optimally tuned RCT-only class. Passing from this conditional statement to expected-training risk additionally requires uniform integrability. Strict asymptotic improvement requires a positive oracle gap that dominates the validation and drift remainders. No part of this argument gives universal finite-sample or realized-sample dominance.

Why λ_Q^* need not optimize sieve risk. Consider an unpenalized linear sieve fitted on RCT pseudo-outcomes with $\omega = 0$, basis b_p , population Gram matrix $H_p = \mathbb{E}_R\{b_p(X)b_p(X)^\top\}$, projection t_p of τ , and leverage $\ell_p(X) = b_p(X)^\top H_p^{-1} b_p(X)$. Under the usual fixed-sieve least-squares expansion, the λ -dependent part of the leading integrated estimation variance is

$$\frac{1}{n} \mathbb{E}_R [\ell_p(X) \text{Var}_R(Z_\lambda | X)]. \quad (49)$$

The oracle λ_Q^* minimizes the same expression without $\ell_p(X)$. The two coefficients coincide when the leverage is constant, or under another condition that makes the two weighted quadratics share a minimizer, but they need not coincide in general. Penalties, ω , and data-adaptive basis selection create further differences between the integrated-noise oracle and the final learner-risk oracle.

For $0 \leq \omega < 1$, define $\tilde{Z}_{\lambda,\omega} = (Z_\lambda - \omega g)/(1 - \omega)$. Completing the square gives

$$\hat{L}_{\text{DR}}(t; \lambda, \omega) = (1 - \omega) \mathbb{P}_{R,n} \{t(X) - \tilde{Z}_{\lambda,\omega}\}^2 \quad (50)$$

$$+ \omega \mathbb{P}_{O,N} \{r(X)(g(X) - t(X))^2\} + \text{a term constant in } t. \quad (51)$$

This form is a computational device for convex squared-loss learners. The original objective must be used at $\omega = 1$, outside $[0, 1]$, or when a nonlinear parameterization makes convexity unclear.

4.2. Implementable F-Learner and prospective R-loss extension

We use *F-Learner* as a provisional working label only for CATE estimators that minimize Equation (28), which is built from the RCT-valid AIPW pseudo-outcome Z_λ . The label does not claim name novelty and does not denote a generic unification of doubly robust and R losses.

For a linear sieve $t_\beta(x) = b(x)^\top \beta$, with $b(x) \in \mathbb{R}^p$, add the penalty $\rho \beta^\top P \beta$, where $\rho \geq 0$ and $P = P^\top \succeq 0$ belongs to $\mathbb{R}^{p \times p}$. Define

$$H_R = \mathbb{P}_{R,n}\{b(X)b(X)^\top\}, \quad H_{O,r} = \mathbb{P}_{O,N}\{r(X)b(X)b(X)^\top\}, \quad (52)$$

$$h_\lambda = \mathbb{P}_{R,n}\{b(X)Z_\lambda\}, \quad d_g = \mathbb{P}_{O,N}\{r(X)b(X)g(X)\} - \mathbb{P}_{R,n}\{b(X)g(X)\}. \quad (53)$$

Whenever the matrix is invertible, the penalized F-Learner solution is

$$\hat{\beta}_{\lambda,\omega} = \{(1-\omega)H_R + \omega H_{O,r} + \rho P\}^{-1} \{h_\lambda + \omega d_g\}. \quad (54)$$

Because h_λ is affine in λ , a fixed- ω candidate path can be computed efficiently. A frozen neural basis replaces $b(x)$ by a representation learned without the candidate-training or validation observations and inherits Equation (54) conditional on that representation. End-to-end neural fitting is nonconvex, and the present linear-sieve guarantees do not cover it.

A prospective R-loss extension can be implemented as an empirical candidate, but it requires a separate theorem before it belongs to the F-Learner class. Let m be a generic fixed, cross-fitted marginal-outcome regression trained independently of the observations on which the following loss is evaluated. One candidate objective is

$$\hat{L}_R(t; \omega) = \mathbb{P}_{R,n}[\{Y - m(X)\} - \{A - e(X)\}t(X)]^2 \quad (55)$$

$$+ \omega \left[\mathbb{P}_{O,N}\{r(X)(g(X) - t(X))^2\} - \mathbb{P}_{R,n}\{(g(X) - t(X))^2\} \right], \quad (56)$$

Randomized treatment implies

$$\mathbb{E}_R[(A - e(X))\{Y - m(X)\} \mid X] = e(X)\{1 - e(X)\}\tau(X) \quad (57)$$

for every fixed m . This identity motivates the residualized objective. No λ channel has yet been derived for this R-loss candidate. A complete result must still derive the precise target under penalties and restricted learners, a principled role for λ , an honest validation score, and estimated-nuisance rates. We therefore present Equation (56) as a planned extension rather than a proved contribution.

Under covariate shift, an exact ratio can be recovered from domain-classification odds. If $S = 1$ denotes RCT membership, $\pi = P(S = 1)$, and $s(x) = P(S = 1 \mid X = x)$, then

$$r(x) = \frac{1 - \pi}{\pi} \frac{s(x)}{1 - s(x)}. \quad (58)$$

An estimated ratio $\hat{r}$ replaces exact target preservation with an explicit drift. Clipping changes the exact transported target and must be reported.

A loss-targeted alternative chooses a nonnegative, out-of-sample-evaluable weight function w to balance

$$\mathcal{G}(\mathcal{T}; g) = \{(g - t)^2 : t \in \mathcal{T}\}. \quad (59)$$

If $\mathbb{E}_O\{w(g - t)^2\} = \mathbb{E}_R\{(g - t)^2\}$ for every $t \in \mathcal{T}$, Equation (30) remains exact with w in place of r . For a linear sieve, balancing bg and $bb^\top$ preserves the minimizer; balancing g^2 is additionally required only to preserve the absolute loss value. Exact balancing on the same observations used to

fit t makes the correction zero or constant in the sieve coefficient. Weight learning and candidate fitting must therefore use different honest blocks or appropriate outer folds.

For approximate weights, define

$$\Delta_{\mathcal{T}}(w) = \sup_{t \in \mathcal{T}} \left| \mathbb{E}_O\{w(X)(g(X) - t(X))^2\} - \mathbb{E}_R\{(g(X) - t(X))^2\} \right|. \quad (60)$$

The uniform population-loss drift is bounded by $|\omega|\Delta_{\mathcal{T}}(w)$, and the excess target risk of the weighted population minimizer is at most $2|\omega|\Delta_{\mathcal{T}}(w)$ when both minimizers exist.

Current scope and open theory. The proved CATE statements cover the AIPW-pseudo-outcome squared loss with fixed honest nuisances and either a common covariate marginal or exact transport. The linear-sieve solution and frozen-feature extension are conditional results. Estimated-ratio rates, adaptive loss-targeted balancing, end-to-end neural optimization, the R-loss extension, and cross-fitted inference remain open theory. Honest splitting prevents direct reuse but does not by itself close these gaps.

5. Experiments

5.1. Completed ATE benchmarks

The completed experiments evaluate ATE estimation only. All nested benchmarks use 20 Monte Carlo repetitions per cell, an RCT-only AIPW baseline, a λ -only outcome-model fusion estimator, an ω -only PPI correction estimator, and the joint estimator. The code retains two stale internal names: `haipw_only` denotes the λ -only estimator and `ppi_power_tuned_only` denotes the ω -only estimator. The first label does not identify the method with the separate HAIPW literature.

The fixed-OBS-budget benchmark contains 16 design cells: three arbitrary nonlinear SCMs and one STAR real- X construction, each under a common covariate marginal and covariate shift and $n_R \in \{20, 30\}$, with $N_O = 5000$. The joint estimator has the smallest RMSE in 11 cells, and the λ -only estimator is best in the remaining five. The joint estimator has lower RMSE than AIPW in all 16 cells, with mean RMSE ratio 0.648 and mean empirical-variance ratio 0.439 relative to AIPW. The λ -only estimator also beats AIPW in all 16 cells, with mean RMSE ratio 0.679. The ω -only estimator is not best in any cell and has mean RMSE ratio 1.005.

The proportional-budget benchmark contains 32 cells with $n_R \in \{20, 30, 50, 100\}$ and $N_O = 100n_R$. The joint estimator has the smallest RMSE in 18 cells, the λ -only estimator in 10, the ω -only estimator in one, and AIPW in three. The joint estimator has lower RMSE than AIPW in 27 of 32 cells, with mean RMSE ratio 0.760 and mean empirical-variance ratio 0.625. The λ -only estimator beats AIPW in 28 cells and has mean RMSE ratio 0.769. The ω -only estimator beats AIPW in 10 cells and has mean RMSE ratio 1.020. These counts show that joint tuning is often useful but is not uniformly best after coefficient estimation.

The STAR benchmark uses only real kindergarten covariates from Project STAR. Treatment assignments, latent confounding, potential outcomes, and effect heterogeneity are generated synthetically. It is therefore a real- X , semi-synthetic benchmark rather than an analysis of STAR’s original intervention or outcomes. The planned WHI CT+OS analysis has not been run because participant-level

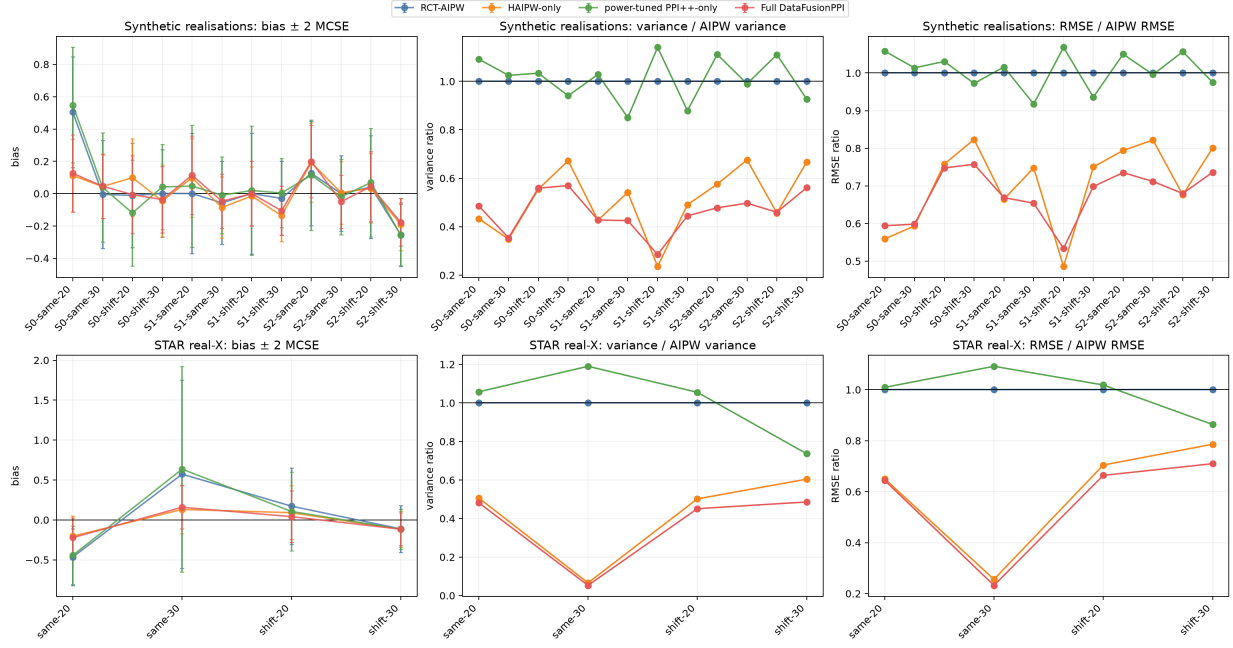


Figure 1: Fixed-OBS-budget ATE benchmark. The composite artifact reports multiple method and regime panels side by side for $n_R \in \{20, 30\}$ and $N_O = 5000$.

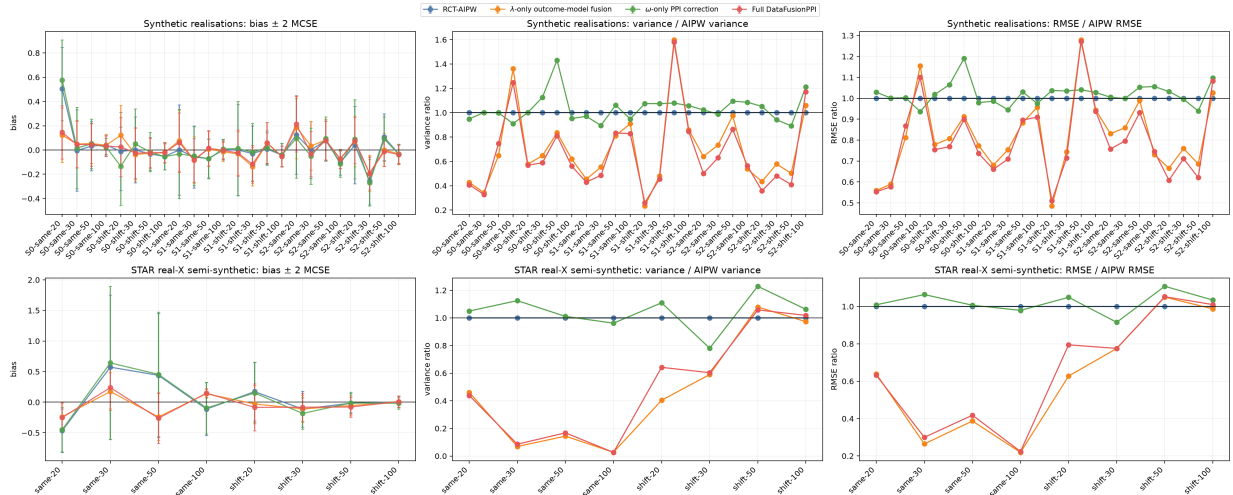


Figure 2: Proportional-budget ATE benchmark. The composite artifact reports multiple method and regime panels side by side for $n_R \in \{20, 30, 50, 100\}$ and $N_O = 100n_R$.

access, a semantic crosswalk, and a survival or censoring extension are still required.

5.2. Planned CATE evaluation and interpretation

CATE experiments are planned. The primary design will compare the proved AIPW-pseudo-outcome F-Learner with the prospective R-loss extension using additive cubic B-splines, frozen neural bases, and linear sieves under a common covariate marginal and covariate shift. Each study will compare trial-only, λ -only, ω -only, and joint tuning using held-out RCT-population CATE risk. Classifier-only, balancing-only, hybrid, and oracle transport weights will be separated. No CATE result is claimed from Figures 1 and 2.

The empirical conclusion is deliberately limited. The two-channel population oracle weakly improves the ATE variance objective when the trial-only coefficient is feasible, but estimated coefficients do not inherit a universal no-harm property. The present 20-repetition studies suggest that the λ channel is the more stable source of gain and that joint tuning can add value in some designs. More Monte Carlo repetitions, completed CATE experiments, additional real- X supports, and an authorized real-outcome analysis are required before making a stronger empirical claim.

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A. Simulation Details

All completed simulation artifacts were generated with base seed 190602. The primary nested benchmarks use 20 outer Monte Carlo repetitions per design cell and compare four estimators: RCT-only AIPW, λ -only outcome-model fusion, ω -only PPI correction, and joint (λ, ω) fusion. The implementation retains the historical internal codes `haipw_only` and `ppi_power_tuned_only` for the second and third estimators, respectively. These codes are not the user-facing method names, and the first does not refer to the separate HAIPW paper.

The arbitrary simulation family uses three nonlinear structural causal models. In every model, a domain indicator S distinguishes the observational sample ($S = 0$) from the randomized sample ($S = 1$); covariates depend on latent variables, observational treatment depends on both measured covariates and an unmeasured confounder, trial treatment follows the known randomized propensity, and outcomes may depend nonlinearly on covariates, treatment, and the latent confounder. The common-marginal regime uses $P_R^X = P_O^X$. The shifted regime changes the covariate distribution while preserving overlap and estimates the density ratio from a domain classifier. The same estimators and evaluation logic are used in both regimes.

The fixed-OBS-budget experiment has 16 cells: four benchmark families, two covariate regimes, and $n_R \in \{20, 30\}$, with $N_O = 5000$. The proportional experiment has 32 cells: four benchmark families, two regimes, and $n_R \in \{20, 30, 50, 100\}$, with the exact mapping $N_O = 100n_R \in \{2000, 3000, 5000, 10000\}$. Coefficients are estimated by the nested honest procedure; no oracle coefficient or oracle density ratio is used in the reported method comparison. The replication-level CSV files, summaries, and composite plots in `materials/` are the canonical numerical artifacts.

The fourth benchmark family reuses the empirical covariate support from the Tennessee Student/Teacher Achievement Ratio (STAR) experiment. Treatment, latent confounding, outcomes, and treatment-effect heterogeneity are generated by the benchmark code. Consequently this is a real- X , semi-synthetic design, not an evaluation of the original STAR outcomes or intervention effect.

Figure 3 records the earlier two-method pilot. It is retained for provenance but is not the basis for the four-method conclusions in Section 5.

No participant-level WHI CT+OS analysis is included. Such an analysis requires authorized data access, a verified trial-observational semantic crosswalk, a prespecified estimand, and methods appropriate for censoring if the selected outcome is time-to-event. No completed CATE experiment or real-outcome result is represented by the present artifacts.

Illustrative random-SCM debugging pilot (20 outer refits per cell)
per stage: RCT n / OBS 5000; total generated per replicate: $3n / 15000$

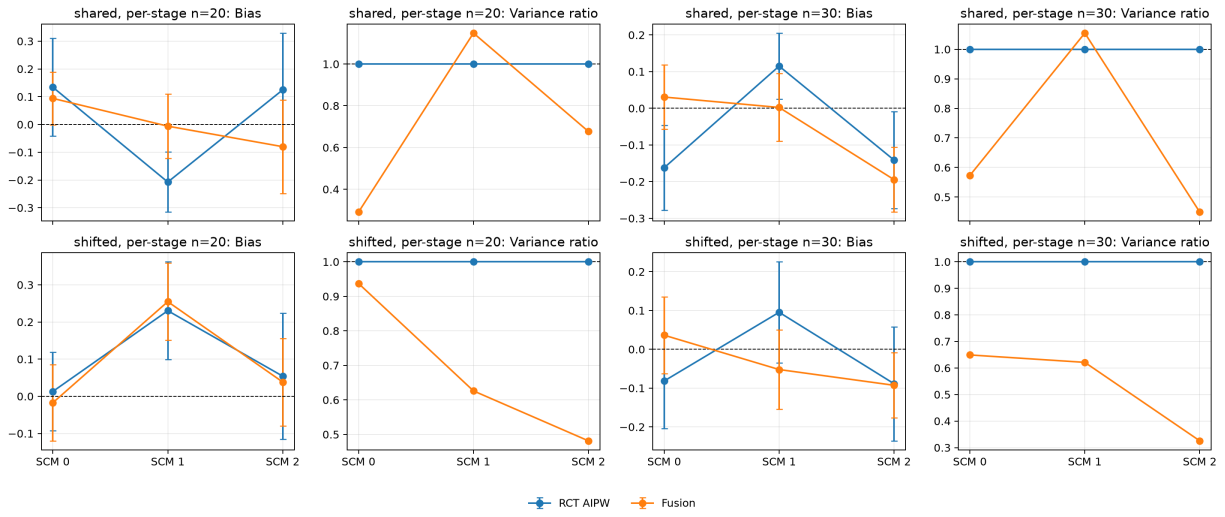


Figure 3: Earlier SSEM ATE pilot. This artifact predates the nested four-method comparison and should be interpreted only as a design and implementation check.

B. Proof Details

B.1. Conditional validity of the trial pseudo-outcome

Fix $q = (q_0, q_1)$ and condition on X . Under trial randomization and consistency,

$$\mathbb{E}_R \left[\frac{A}{e(X)} \{Y - q_1(X)\} \mid X \right] = \mathbb{E}_R \{Y(1) - q_1(X) \mid X\}, \quad (61)$$

$$\mathbb{E}_R \left[\frac{1 - A}{1 - e(X)} \{Y - q_0(X)\} \mid X \right] = \mathbb{E}_R \{Y(0) - q_0(X) \mid X\}. \quad (62)$$

Substituting these two equalities into the definition of φ cancels the two fitted regressions and yields $\mathbb{E}_R \{\varphi(V; q) \mid X\} = \tau(X)$. Applying the identity to $q = \hat{\mu}_R$, $q = \hat{\mu}_O$, and their affine blend gives

$$\mathbb{E}_R(Z_\lambda \mid X) = \tau(X), \quad \mathbb{E}_R(\Delta \mid X) = 0. \quad (63)$$

These statements are conditional on the pre-evaluation sigma-field so that the fitted functions are fixed.

B.2. Proof of Theorem 1

Using $\mathbb{E}_R(Z_\lambda) = \theta$ and exact transport,

$$\mathbb{E}_O\{r(X)g(X)\} = \mathbb{E}_R\{g(X)\}, \quad (64)$$

the conditional mean of Equation (10) is θ . Rewrite the centered estimator as

$$\hat{\theta}_r(\lambda, \omega) - \theta = (\mathbb{P}_{R,n} - P_R)\{Z_\lambda - \omega g\} + (\mathbb{P}_{O,N} - P_O)\{\omega r g\}. \quad (65)$$

The two summands are independent conditional on $\mathcal{F}$. The variance of an empirical mean of m independent observations is the population variance divided by m , which proves Equation (12). Unbiased sample variances on the two independent evaluation blocks therefore give the displayed fixed-selected-coefficient variance estimator in Section 3. Coverage of its Wald interval additionally uses the stated conditional Lindeberg and moment assumptions; variance unbiasedness alone is not a central limit theorem.

If an estimated ratio $\hat{r}$ replaces r , the conditional mean changes by exactly

$$\omega P_O\{(\hat{r} - r)g\}. \quad (66)$$

Thus honest evaluation does not restore exact unbiasedness. Negligibility of both the root- n drift and the OBS empirical remainder is required before the fixed-ratio evaluation variance can be used as a first-order variance formula.

B.3. Proof of Theorem 2

Expand the first variance in Equation (12) using $Z_\lambda = Z_0 + \lambda\Delta$. After collecting terms and multiplying by n , the coefficient-dependent part is

$$2C\lambda - 2D\omega + A\lambda^2 - 2E\lambda\omega + B\omega^2, \quad (67)$$

which is $2\ell^\top\beta + \beta^\top\mathbf{H}\beta$. The representation

$$\begin{pmatrix} u & v \end{pmatrix} \mathbf{H} \begin{pmatrix} u \\ v \end{pmatrix} = \text{Var}_R(u\Delta - vg) + \frac{n}{N}v^2 \text{Var}_O(rg) \quad (68)$$

proves positive semidefiniteness. If $AB - E^2 > 0$, the first-order condition $\mathbf{H}\beta = -\ell$ has the unique solution

$$\lambda^\star = \frac{ED - BC}{AB - E^2}, \quad \omega^\star = \frac{AD - EC}{AB - E^2}. \quad (69)$$

When $\mathbf{H}$ is singular, nonnegativity of the variance quadratic implies $\ell \in \text{Range}(\mathbf{H})$; otherwise a null-space direction would make the expression arbitrarily negative. The complete minimizer set is therefore $-\mathbf{H}^+\ell + \text{Null}(\mathbf{H})$. Continuity of the quadratic proves attainment on every prespecified compact feasible set.

Finally, because g is X -measurable,

$$E = \mathbb{E}_R[\{g(X) - P_Rg\}\mathbb{E}_R(\Delta \mid X)] = 0, \quad (70)$$

$$D = \text{Cov}_R\{\mathbb{E}_R(Z_0 \mid X), g(X)\} = \text{Cov}_R\{\tau(X), g(X)\}. \quad (71)$$

Substitution yields Equation (21). Since the population objective is minimized over a set containing $(0, 0)$, its oracle value cannot exceed the trial-only value. This comparison concerns the population objective, not coefficients estimated from a finite tuning sample.

B.4. Prediction-powered CATE risk identity

Fix t and condition on X . Since $\mathbb{E}_R(Z_\lambda - \tau(X) \mid X) = 0$,

$$\mathbb{E}_R\{Z_\lambda - t(X)\}^2 = \mathbb{E}_R\{Z_\lambda - \tau(X)\}^2 + \mathbb{E}_R\{t(X) - \tau(X)\}^2. \quad (72)$$

Exact transport also gives

$$\mathbb{E}_O\{r(X)(g(X) - t(X))^2\} = \mathbb{E}_R\{(g(X) - t(X))^2\}. \quad (73)$$

The correction in Equation (28) therefore has mean zero, which proves Equation (30). In particular, population expected loss is flat in ω . The part depending on λ is

$$Q(\lambda) = Q(0) + 2C\lambda + A\lambda^2, \quad (74)$$

so when $A > 0$ its minimizer is $-C/A$, the same integrated-noise oracle as in the population-orthogonal ATE problem. If $A = 0$, then $\Delta = 0$ almost surely by conditional mean zero, and Q is flat in λ .

For a fixed t and λ , the empirical loss is the sum of independent empirical means of $U_{t,\lambda} - \omega G_t$ under the trial and $\omega r G_t$ under the observational law. Expanding its variance as a quadratic in ω and differentiating gives Equation (32). Because t is fixed in this calculation, the formula does not identify the coefficient minimizing the risk of a learner chosen after seeing the same loss.

For the linear sieve, expand Equation (28) in β , add $\rho\beta^\top P\beta$, and set the gradient to zero. The resulting normal equation is

$$\{(1 - \omega)H_R + \omega H_{O,r} + \rho P\}\beta = h_\lambda + \omega d_g, \quad (75)$$

which proves Equation (54) whenever the matrix is invertible.

B.5. Loss-targeted balance and approximate drift

Suppose a nonnegative out-of-sample-evaluable weight w satisfies

$$P_O\{w(g - t)^2\} = P_R\{(g - t)^2\} \quad \text{for every } t \in \mathcal{T}. \quad (76)$$

Replacing r by w then leaves the population correction exactly zero for every candidate, so the target identity remains valid. If $t_\beta = b^\top \beta$, expanding

$$(g - b^\top \beta)^2 = g^2 - 2\beta^\top b g + \beta^\top b b^\top \beta \quad (77)$$

shows that balancing bg and $bb^\top$ preserves the minimizer. Balancing g^2 is needed only to preserve the absolute loss value.

For approximate weights, the difference between the weighted and target population losses is bounded uniformly over $\mathcal{T}$ by $|\omega|\Delta_{\mathcal{T}}(w)$. Let t_w minimize the weighted population loss and t_0 minimize the target loss. Applying the uniform bound at both minimizers gives

$$L_0(t_w) \leq L_w(t_w) + |\omega|\Delta_{\mathcal{T}}(w) \quad (78)$$

$$\leq L_w(t_0) + |\omega|\Delta_{\mathcal{T}}(w) \quad (79)$$

$$\leq L_0(t_0) + 2|\omega|\Delta_{\mathcal{T}}(w), \quad (80)$$

which is the claimed excess-risk bound. If the same finite sample is used both to impose exact sieve balance and to fit β , the balanced linear and quadratic coefficient terms cancel from the empirical correction (and the constant cancels as well if g^2 is balanced). This algebra explains why weight learning and candidate fitting require honest separation or a separate outer-fold argument.